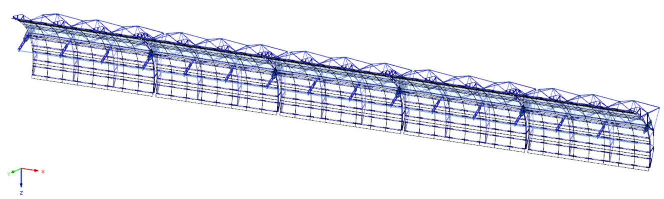
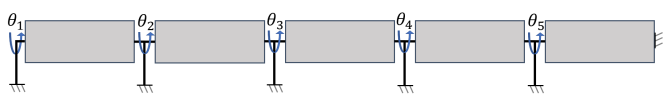
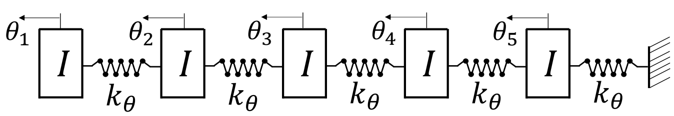
Overview. My teaching style resembles my research style: I weave the **technological, physical and mathematical** strands to introduce problems in all their complexity. I cherish science outreach, and, to me, the first place to do so is the classroom. Teaching and mentoring help deepen my understanding of concepts, and thus they **foster my research**. I can proudly say that a publication of mine [1] grew from a question posed by an outstanding student: *what does "resonance" really mean?* I encourage interactions grounded in mutual respect and a shared passion for science, **supporting students from diverse backgrounds** by ensuring their perspectives help shape how we approach the material.



(a) FEM model of a five-module parabolic solar collector



(b) Scheme highlighting torsion degrees of freedom



(c) Five-dof lumped torsional vibration model

FIGURE 1. Solar collectors as a dynamic torsional system excited by wind loads (angle θ, polar moment of inertia I, and module torsional spring k_θ).

Teaching and mentoring experience. I acted as **teaching assistant** (TA) during the academic years 2017-18, 18-19 and 19-20, in the context of AM/CE 151a **Dynamics and Vibrations**. Prof. Asimaki entrusted me with daily operations (grading, office hours, recitations and exercise sessions) and occasional lectures. My enthusiasm was rewarded in the feedback reports (I was recognized with the campus-wide "**best TA distinction**" in the fall term 17-18). During the spring semester 21-22, I collaborated with Prof. Molinari in the context of CIVIL-422 **Advanced Continuum Mechanics** (an introduction to finite-kinematics elasticity and to constitutive modeling) preparing lectures, holding office hours and occasionally teaching when he was on travel. Again, during the spring semester 23-24, I aided him and Prof. Lecampion with a new class of similar scope, CIVIL-425 **Continuum Mechanics and Applications**. I served as the primary instructor in the spring semester 24-25, receiving positive reviews (see attached CV). At EPFL, I have supervised ten semester projects (bachelor and master) and two master theses. My first PhD student, Mr. Gaëtan Cortes, passed his candidacy exam in October 2025.

Teaching methods. I lecture using the board and emphasizing **threshold concepts** [2] openly, thus trying to ensure that the foundational concepts are assimilated before building upon. This strategy leads to a continuous expansion of students' **zone of proximal development** [3]; the goal is that they can continue delving in the subject as independent learners. I look for inspiration in my **own experience**. E.g., Figure 1 shows a modular trough collector — a system I once designed in **industry** — known to display torsional vibrational modes when excited by wind, which made for an exam exercise.

New classes. At the **graduate level**, I would like to eventually prepare two new one-semester classes. "**Fluid-Solid Interactions at Low Reynolds Numbers**" would provide a broad introduction to problems in which solid deformation and thin fluid layers are strongly coupled; it would cover both the continuum mechanics fundamentals together with selected applications, e.g., elastohydrodynamic lubrication [4] and haptics [5]. "**Group Theory for Engineers**" would introduce basic notions of group theory [6] (group axioms, isomorphisms, symmetries) while illustrating these concepts with examples, e.g., 3D space rotations using $SO(3)$ and transfer matrices for wave propagation as elements of $SL_2(\mathbb{R})$ [7].

References

[1] **Garcia-Suarez, J.**, Asimaki, D. (2020) "On the fundamental resonance mode of inhomogeneous soil deposits". *Soil Dynamics and Earthquake Engineering*.

[2] Meyer, J., Land, R. (2003) "Threshold concepts and troublesome knowledge: linkages to ways of thinking and practising within the disciplines".

[3] Fila, N., Fernandez, T., Purzer, S., Bohlin, A. (2016) *Innovation and the zone of proximal development in engineering education*. In: 2016 ASEE Annual Conference & Exposition.

[4] Dowson, D., and Higginson, G. R. (2014) *Elasto-Hydrodynamic Lubrication*. Publisher: Pergamon.

[5] Wiertlewski, M., Fenton Friesen, R., and Colgate, J. E. (2016) "Partial squeeze film levitation modulates fingertip friction", *Proceedings of the National Academy of Sciences*

[6] Artin, M. (2011) *Algebra (2nd edition)*. Publisher: Prentice Hall.

[7] **Garcia-Suarez, J.** (2022) "Harmonic decomposition of the trace of 1D transfer matrices in layered media", *Journal of the Mechanics and Physics of Solids*.